Project Proposal: Variational Autoencoder (VAE) for Unsupervised Financial Anomaly Detection

Course: IT429 Generative AI (B.Tech. Information Technology)
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1. Project Objective The objective of this project is to design, train, and evaluate a Variational Autoencoder (VAE) from scratch to detect anomalous and potentially fraudulent financial transactions. Traditional discriminative models often fail in fraud detection due to extreme class imbalance, as fraudulent records represent a microscopic fraction of overall data. This project solves that problem through a generative approach: by training a VAE exclusively on normal transaction data, the network learns the exact probability distribution of legitimate behavior. Irregular transactions are then mathematically flagged based on their inability to be accurately reconstructed by the model.

2. Syllabus Alignment This project strictly adheres to the core concepts outlined in the course structure:

- **Unit 1 (Introduction to Generative AI):** Practical application of latent spaces and probability distributions, demonstrating a clear use-case where generative modeling outperforms discriminative modeling.
- **Unit 2 (Foundations of Generative Models):** Full mathematical and programmatic implementation of a Variational Autoencoder, focusing on the architecture, the reparameterization trick, and custom loss functions.

3. Methodology and System Architecture The project abandons high-level API wrappers in favor of coding a custom generative neural network. The pipeline consists of:

- **The Encoder Network:** Compresses high-dimensional tabular data (e.g., transaction amount, time, location, frequency) into a lower-dimensional latent space, outputting mean (μ) and variance (σ^2) vectors.
- **Latent Space Sampling:** Implements the reparameterization trick to sample from the learned distribution while maintaining a differentiable pipeline for backpropagation.
- **The Decoder Network:** Reconstructs the original input features from the sampled latent vector.
- **Anomaly Scoring (Inference):** The model is trained to minimize a custom objective function combining Mean Squared Error (Reconstruction Loss) and Kullback-Leibler (KL) Divergence. During testing, anomalous data will fall outside the learned latent space,

causing a statistically significant spike in reconstruction loss. A calculated threshold will trigger the fraud alert.

4. Achievability and Implementation Plan This project is highly feasible for a group of 3 students to complete within the semester at zero financial cost.

- **Data Source:** Open-source, highly imbalanced datasets, such as the Kaggle Credit Card Fraud Detection dataset, which provides robust tabular features for training.
- **Technology Stack:** Python, PyTorch (or TensorFlow) for building the VAE classes, and Pandas/NumPy for data preprocessing.
- **Compute Resources:** Model training and validation will be executed entirely on Google Colab utilizing free cloud GPUs.
- **Final Deliverable:** A comprehensively documented Jupyter Notebook containing the data ingestion pipeline, custom neural network classes, the training loop, and evaluation metrics (Precision/Recall curves) proving the efficacy of the anomaly threshold.